

# Brock Allan

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## EDUCATION

University of California, Irvine

Irvine, California

Bachelor's of Science in **Computer Science**

Cumulative GPA: **3.6**.

- **Coursework:** Systems & Algorithms, **Digital Logic**, Machine Learning, Multivariable Calculus, Linear Algebra, Data Structure Implementation & Analysis, Computer Vision
- **Distinctions:** CLAP certified, **Chancellor's Excellence** Scholar, Dean's Honor List.

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## SKILLS & TOOLS

- Coding **Languages:** Java, **Python**, C++, **R**, **Assembly**, C, C#, HTML, Lisp, Prolog
- Industry **Software:** **Unreal Engine 5**, **Unity**, Photoshop, Google & Excel Spreadsheets
- Leadership **Positions:** FFA President, NHS President, MESA Member, **Project Leader**

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## TECHNICAL EXPERIENCE

Web Search Engine & Crawler

|January 2026 - March 2026|

- Architected a web search engine and **multi-threaded crawler** from scratch in **Python**, achieving **sub-300ms** query response times from datasets of **50,000+ web pages**.
- Engineered crawling logic that strictly adhered to **Robots.txt politeness policies** and utilized **frontier-management** strategies to avoid crawler **traps** and **dead-end** pages.
- Developed an optimized **scraper** to target and extract **high-density textual information**, ensuring quality for the **search index** under strict operational and **memory constraints**.

Custom Operating System

|September 2025 - January 2026|

- Developed a custom **Unix-like shell** for **x86/RISC-V** in **C** including a **recursive parser** that handles multi-stage **pipelines**, **I/O redirection**, and **comment-aware grammar** for **automated script execution**.
- Created several **kernel level system calls** from scratch including **ps** to monitor process lifecycle states such as **Running**, **Sleeping**, and **Zombie**.
- Expanded **exec** to identify **non-ELF text scripts** and automatically re-route execution through the designated **shell interpreter**.

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## PROFESSIONAL EXPERIENCE

Computer Consultant & Data Analyst - Freedom Tax Insurance

|May 2020 - June 2024|

- Identified **hardware** issues caused by component failures, modeled using **R**.
- Used **diagnostic** tools to repair and refurbish company devices and **servers**.
- Obtained experience in creating professional **reports** that identified company issues.